Second-Order Perturbative Expansion of a Layered Clifford Circuit with Small Unitary Insertions

Notes (self-contained)

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1 Problem setup

Consider an n-qubit layered circuit of depth L whose ideal evolution is a product of layer unitaries

$$U \;=\; \prod_{l=1}^{L} u_l, \tag{1}$$

where each u_l is a Clifford unitary (the Clifford assumption is only needed for later applications; the algebra below holds for general unitaries). We use the convention that the product is time-ordered from early to late: the factor with smaller l is earlier in time.

We perturb each layer by inserting a small unitary generated by a Hermitian operator H_l:

$$u_l' \;=\; e^{-i\epsilon_l H_l}\, u_l, \qquad |\epsilon_l| \ll 1, \qquad H_l = H_l^\dagger. \tag{2}$$

The perturbed circuit is therefore

$$U' \;=\; \prod_{l=1}^{L} u_l' \;=\; \prod_{l=1}^{L} \Big(e^{-i\epsilon_l H_l} u_l\Big). \tag{3}$$

Goal. Derive (i) a second-order expansion of U' in $\{\epsilon_l\}$, and (ii) a second-order expansion of the corresponding unitary channel

$$\mathcal{E}_L(\rho) \;=\; U' \rho U'^\dagger, \tag{4}$$

in a form that a reader can replicate.

2 Notation for partial products

Define partial products of the ideal layer unitaries:

$$U_{a:b} \;:=\; \prod_{l=a}^{b} u_l, \qquad (a \le b), \tag{5}$$

and adopt the convention that an empty product equals the identity:

$$U_{a:b} = \mathbb{I} \quad \text{if } a > b.. \tag{6}$$

Then $U = U_{1:L}$. For each layer l,

$$U_{l+1:L} = \prod_{j=l+1}^{L} u_j, \qquad U_{1:l} = \prod_{j=1}^{l} u_j. \tag{7}$$

3 Pushing insertions to the circuit end

Conjugate each insertion $e^{-i\epsilon_l H_l}$ through the *later* ideal evolution. Define the "Heisenberg-pushed" generators

$$\widetilde{H}_l \;:=\; U_{l+1:L}\, H_l\, U_{l+1:L}^{\dagger}. \tag{8}$$

Then one can rewrite the perturbed circuit (3) as

$$\boxed{U' \;=\; \Big(\prod_{l=1}^{L} e^{-i\epsilon_l \widetilde{H}_l}\Big)\, U.} \tag{9}$$

Derivation of (9). Use the identity $e^{-i\epsilon H} W = W\, e^{-i\epsilon W^{\dagger} H W}$ for any unitary W, applied iteratively with $W = U_{l+1:L}$. This replaces each H_l by $\widetilde{H}_l$ as in (8), and factors out the ideal product $U = U_{1:L}$ on the right, giving (9). □

Remark (Clifford specialization). If each u_l is Clifford and each H_l is a Pauli (or low-weight sum of Paulis), then $\widetilde{H}_l$ is again a Pauli (or the conjugated low-weight sum).

4 Second-order expansion of the perturbed unitary

Define

$$K \;:=\; \prod_{l=1}^{L} e^{-i\epsilon_l \widetilde{H}_l}, \qquad \text{so that } \; U' = KU.$$

Using

$$e^{-i\epsilon_l \widetilde{H}_l} = \mathbb{I} - i\epsilon_l \widetilde{H}_l - \frac{\epsilon_l^2}{2}\widetilde{H}_l^2 + O(\epsilon_l^3), \tag{10}$$

and keeping terms up to total degree 2 in the set $\{\epsilon_l\}$, we obtain

$$\boxed{U' \;=\; \Big[\mathbb{I} - i\sum_{l=1}^{L}\epsilon_l \widetilde{H}_l - \frac{1}{2}\sum_{l=1}^{L}\epsilon_l^2 \widetilde{H}_l^2 - \sum_{1\le i<j\le L}\epsilon_j\epsilon_i\, \widetilde{H}_j \widetilde{H}_i\Big]\, U \;+\; O(\epsilon^3).} \tag{11}$$

Here $O(\epsilon^3)$ denotes terms of total degree at least 3 in the ϵ_l's. The cross-term is time-ordered: $j > i$ implies $\widetilde{H}_j$ appears to the left of $\widetilde{H}_i$.

5 Second-order expansion of the unitary channel

Define the ideal-output state

$$\rho_0 \;:=\; U\rho U^{\dagger}. \tag{12}$$

Then the perturbed channel (4) becomes

$$\mathcal{E}_L(\rho) = U'\rho U'^{\dagger} = K\rho_0 K^{\dagger}. \tag{13}$$

Write a degree expansion of K:

$$K = \mathbb{I} + A + B + O(\epsilon^3), \tag{14}$$

where the first- and second-order parts are

$$A = -i \sum_{l=1}^{L} \epsilon_l \widetilde{H}_l, \tag{15}$$

$$B = -\frac{1}{2} \sum_{l=1}^{L} \epsilon_l^2 \widetilde{H}_l^2 - \sum_{1 \le i < j \le L} \epsilon_j \epsilon_i \, \widetilde{H}_j \widetilde{H}_i. \tag{16}$$

Since each $\widetilde{H}_l$ is Hermitian, $A^\dagger = -A$.

Substitute (14) into (13) and keep terms up to $O(\epsilon^2)$:

$$K \rho_0 K^\dagger = \rho_0 + A\rho_0 + \rho_0 A^\dagger + B\rho_0 + \rho_0 B^\dagger + A\rho_0 A^\dagger + O(\epsilon^3). \tag{17}$$

5.1 First order

Using $A^\dagger = -A$,

$$\boxed{\mathcal{E}_L(\rho) = \rho_0 - i \sum_{l=1}^{L} \epsilon_l \, [\widetilde{H}_l, \rho_0] + O(\epsilon^2).} \tag{18}$$

5.2 Second order

Separate the $O(\epsilon^2)$ correction into (i) "self" terms $\propto \epsilon_l^2$ and (ii) cross terms $\propto \epsilon_i \epsilon_j$ with $i < j$.

Self terms. Collect the contributions proportional to ϵ_l^2 from $B\rho_0 + \rho_0 B^\dagger + A\rho_0 A^\dagger$ and simplify:

$$\boxed{\mathcal{E}_L(\rho)\Big|_{\text{self},\, O(\epsilon^2)} = -\frac{1}{2} \sum_{l=1}^{L} \epsilon_l^2 \, [\widetilde{H}_l, [\widetilde{H}_l, \rho_0]].} \tag{19}$$

Cross terms. For $i < j$, the second-order cross contribution can be written in an anticommutator-commutator form:

$$\boxed{\mathcal{E}_L(\rho)\Big|_{\text{cross},\, O(\epsilon^2)} = -\sum_{1 \le i < j \le L} \epsilon_i \epsilon_j \, \{\widetilde{H}_j, \, [\widetilde{H}_i, \rho_0]\},} \tag{20}$$

where $\{A, B\} = AB + BA$.

Full second-order channel. Combining (18), (19), and (20) yields

$$\boxed{\mathcal{E}_L(\rho) = \rho_0 - i \sum_{l=1}^{L} \epsilon_l \, [\widetilde{H}_l, \rho_0] - \frac{1}{2} \sum_{l=1}^{L} \epsilon_l^2 \, [\widetilde{H}_l, [\widetilde{H}_l, \rho_0]] - \sum_{1 \le i < j \le L} \epsilon_i \epsilon_j \, \{\widetilde{H}_j, \, [\widetilde{H}_i, \rho_0]\} + O(\epsilon^3).} \tag{21}$$

6 Replication checklist

A reader can replicate the derivation via:

1. Fix the product convention and define $U_{a:b}$ with the empty-product rule (6).
2. Rewrite U' as $U' = KU$ by conjugating each insertion through the later ideal evolution, giving $\widetilde{H}_l$ in (8).

3. Expand each factor $e^{-i\epsilon_l \widetilde{H}_l}$ to second order and multiply, keeping only terms of total degree ≤ 2, to obtain (11).
4. For the channel, set $\rho_0 = U\rho U^\dagger$ and expand $K\rho_0 K^\dagger$ using (17); simplify using commutator/anticommutator identities to obtain (21).

7 Second-Order Weak-Noise Expansion for Correlated Classical Pauli Hamiltonian Noise

We derive a *replicable* second-order (weak-noise) expression for the *noise-averaged* output density matrix of a layered n-qubit circuit driven by ideal layer unitaries (typically Clifford), in the presence of *correlated classical stochastic Hamiltonian noise* consisting of single-qubit dephasing terms and pairwise correlated dephasing terms.

Output. Given an input state ρ, the ideal circuit output is $\rho_0 = U\rho U^\dagger$. We will show that the *ensemble-averaged* noisy output takes the universal second-order form

$$\boxed{\langle \rho_{\text{out}} \rangle = \rho_0 - i\,[H_{\text{av}}, \rho_0] - \frac{1}{2}\sum_{a,b} \Gamma_{ab}\,[P_a, [P_b, \rho_0]] + O(\text{noise}^3).} \tag{22}$$

Here $\{P_a\}$ is a Pauli-string basis (restricted to the support actually generated by the toggling frame), H_{av} is a (possibly zero) mean drift Hamiltonian, and Γ_{ab} is a coefficient matrix given by double time integrals of the *toggling-frame* noise correlation functions. All steps are spelled out below.

8 Model

8.1 Ideal layered circuit

Let the ideal circuit consist of L instantaneous layer unitaries

$$U = \prod_{l=1}^{L} u_l, \tag{23}$$

with total duration $T = L\tau$, where each layer occupies an interval $I_l = [(l-1)\tau, l\tau)$. (Only the piecewise-constant nature matters; "Clifford" is not required for the algebra, but becomes useful because it maps Paulis to Paulis under conjugation.)

Let $U_0(t)$ denote the ideal (noise-free) evolution up to time t, i.e.,

$$U_0(t) = u_{l(t)} u_{l(t)-1} \cdots u_1 \quad \text{for } t \in I_{l(t)}. \tag{24}$$

Thus $U_0(T) = U$.

8.2 Correlated classical Hamiltonian noise

During layer l (i.e., for $t \in I_l$), the system experiences an additive stochastic Hamiltonian

$$H_e(t) = \sum_{i=1}^{n} \beta_i(t)\, Z_i \; + \sum_{1 \leq i < j \leq n} J_{ij}(t)\, Z_i Z_j. \tag{25}$$

The real-valued processes $\beta_i(t)$ and $J_{ij}(t)$ may be *temporally correlated* and *spatially correlated*:

$$\langle \beta_i(t)\beta_{i'}(t') \rangle - \langle \beta_i(t) \rangle \langle \beta_{i'}(t') \rangle \neq 0, \tag{26}$$

$$\langle J_{ij}(t) J_{i'j'}(t') \rangle - \langle J_{ij}(t) \rangle \langle J_{i'j'}(t') \rangle \neq 0, \tag{27}$$

and similarly for cross-correlations $\langle \beta_i(t) J_{i'j'}(t') \rangle$. Angle brackets $\langle \cdot \rangle$ denote ensemble average.

Weak-noise assumption. We assume amplitudes are small enough that a second-order (Dyson) truncation is accurate:

$$\int_0^T \|H_e(t)\|\, dt \ll 1. \tag{28}$$

8.3 Noisy evolution and the averaged channel

For a given noise realization, the lab-frame time-ordered propagator is

$$U_{\text{tot}}(T) = \mathcal{T} \exp\Big(- i \int_0^T \big(H_{\text{ctrl}}(t) + H_e(t)\big)\, dt\Big), \tag{29}$$

where $H_{\text{ctrl}}(t)$ generates the ideal layer unitaries. We will work in the toggling frame of the ideal evolution, so that only H_e appears.

The corresponding (random) unitary channel is $\rho \mapsto U_{\text{tot}}(T)\rho U_{\text{tot}}^\dagger(T)$. The *physical* noisy channel of interest is the ensemble average

$$\langle \mathcal{E}(\rho)\rangle := \big\langle U_{\text{tot}}(T)\rho U_{\text{tot}}^\dagger(T)\big\rangle. \tag{30}$$

9 Toggling frame reduction

Define the toggling-frame (interaction-picture) noise Hamiltonian

$$\widetilde{H}_e(t) := U_0^\dagger(t)\, H_e(t)\, U_0(t). \tag{31}$$

Then the total propagator factorizes as

$$U_{\text{tot}}(T) = U\, U_I(T), \qquad U_I(T) = \mathcal{T} \exp\Big(- i \int_0^T \widetilde{H}_e(t)\, dt\Big). \tag{32}$$

Hence for any input ρ, letting $\rho_0 := U\rho U^\dagger$,

$$\rho_{\text{out}} = U_{\text{tot}}(T)\rho U_{\text{tot}}^\dagger(T) = U_I(T)\, \rho_0\, U_I^\dagger(T). \tag{33}$$

Therefore it suffices to expand the toggling-frame channel generated by $\widetilde{H}_e(t)$ acting on ρ_0.

Pauli expansion (Clifford case). If $U_0(t)$ is Clifford for all t (instantaneous Clifford layers), then for each t the conjugates $U_0^\dagger(t) Z_i U_0(t)$ and $U_0^\dagger(t) Z_i Z_j U_0(t)$ are Pauli strings (possibly $\pm X, \pm Y, \pm Z$ on each qubit). Thus $\widetilde{H}_e(t)$ can be written as

$$\widetilde{H}_e(t) = \sum_a \xi_a(t)\, P_a, \tag{34}$$

for a (finite) set of Pauli strings $\{P_a\}$ and real stochastic coefficients $\xi_a(t)$ obtained by applying the Clifford "Pauli frame" to the original processes β_i, J_{ij}.

10 Dyson expansion to second order

Let

$$A := \int_0^T \widetilde{H}_e(t)\, dt, \qquad B := \int_0^T dt_2 \int_0^{t_2} dt_1\, \widetilde{H}_e(t_2)\widetilde{H}_e(t_1). \tag{35}$$

The time-ordered exponential admits the Dyson expansion

$$U_I(T) = \mathbb{I} - iA - B + O(\text{noise}^3), \qquad U_I^\dagger(T) = \mathbb{I} + iA - B^\dagger + O(\text{noise}^3), \tag{36}$$

where $B^\dagger = \int_0^T dt_2 \int_0^{t_2} dt_1\, \widetilde{H}_e(t_1)\widetilde{H}_e(t_2)$ because $\widetilde{H}_e(t)$ is Hermitian.

Insert (36) into (33) and keep terms through second order:

$$\begin{aligned}\rho_{\text{out}} &= (\mathbb{I} - iA - B)\,\rho_0\,(\mathbb{I} + iA - B^\dagger) + O(\text{noise}^3)\\ &= \rho_0 - i[A,\rho_0] + \big(A\rho_0 A - B\rho_0 - \rho_0 B^\dagger\big) + O(\text{noise}^3).\end{aligned} \tag{37}$$

11 Ensemble average and the universal second-order form

Take expectation of (37). Define the mean toggling-frame Hamiltonian

$$\widetilde{H}_{\text{av}}(t) := \langle \widetilde{H}_e(t)\rangle, \qquad H_{\text{av}} := \int_0^T \widetilde{H}_{\text{av}}(t)\,dt = \langle A\rangle. \tag{38}$$

Also define fluctuations $\Delta\widetilde{H}_e(t) := \widetilde{H}_e(t) - \langle \widetilde{H}_e(t)\rangle$ and $\Delta A := A - \langle A\rangle$.

11.1 First order

The averaged first-order term is

$$\langle -i[A,\rho_0]\rangle = -i[\langle A\rangle,\rho_0] = -i[H_{\text{av}},\rho_0]. \tag{39}$$

If the noise is zero-mean in the toggling frame (often the case), then $H_{\text{av}} = 0$.

11.2 Second order: double-commutator structure

A standard identity shows that

$$A\rho_0 A - \frac{1}{2}\{A^2,\rho_0\} = -\frac{1}{2}[A,[A,\rho_0]]. \tag{40}$$

Using $B + B^\dagger = A^2$ (the two time-ordered domains partition the square), one finds

$$A\rho_0 A - B\rho_0 - \rho_0 B^\dagger = -\frac{1}{2}[A,[A,\rho_0]]. \tag{41}$$

Therefore, for each realization,

$$\rho_{\text{out}} = \rho_0 - i[A,\rho_0] - \frac{1}{2}[A,[A,\rho_0]] + O(\text{noise}^3). \tag{42}$$

Now average. Write $A = \langle A\rangle + \Delta A$ and keep terms through second order:

$$\boxed{\langle \rho_{\text{out}}\rangle = \rho_0 - i[H_{\text{av}},\rho_0] - \frac{1}{2}\langle[\Delta A,[\Delta A,\rho_0]]\rangle + O(\text{noise}^3).} \tag{43}$$

11.3 Pauli-basis coefficient matrix

Assume the Pauli expansion (34):

$$\widetilde{H}_e(t) = \sum_a \xi_a(t) P_a, \qquad P_a \text{ Pauli strings.}$$

Then

$$\Delta A = \int_0^T \Delta\widetilde{H}_e(t)\,dt = \sum_a \Big(\int_0^T \Delta\xi_a(t)\,dt\Big) P_a. \tag{44}$$

Plugging into (43) gives the main result (22) with

$$\boxed{\Gamma_{ab} = \int_0^T dt \int_0^T dt' \,\Big\langle \Delta\xi_a(t)\,\Delta\xi_b(t')\Big\rangle.} \tag{45}$$

12 How to compute $\xi_a(t)$ from the original processes

In the model (25), each term is either Z_i or Z_iZ_j. Under a (piecewise-constant) Clifford Pauli frame, for each t there exist Pauli strings $P_i(t)$ and $P_{ij}(t)$ such that

$$U_0^\dagger(t)Z_iU_0(t) = P_i(t), \qquad U_0^\dagger(t)Z_iZ_jU_0(t) = P_{ij}(t). \tag{46}$$

Hence

$$\widetilde{H}_e(t) = \sum_i \beta_i(t)\, P_i(t) + \sum_{i<j} J_{ij}(t)\, P_{ij}(t). \tag{47}$$

To match (34), fix a Pauli basis $\{P_a\}$ containing all $P_i(t), P_{ij}(t)$ that occur (over all t). Then $\xi_a(t)$ is obtained by mapping each occurring Pauli string to its basis label a (including the $\pm$ sign), and assigning the corresponding coefficient $\beta_i(t)$ or $J_{ij}(t)$.

13 Replication checklist

A reader can reproduce the result by:

1. Defining $U_0(t)$ and the toggling-frame Hamiltonian $\widetilde{H}_e(t) = U_0^\dagger(t)H_e(t)U_0(t)$.
2. Writing $U_I(T) = \mathcal{T}\exp(-i\int_0^T \widetilde{H}_e(t)\,dt)$ and performing the Dyson expansion to second order (36).
3. Multiplying $(\mathbb{I} - iA - B)\rho_0(\mathbb{I} + iA - B^\dagger)$ and simplifying to obtain (37).
4. Using $B + B^\dagger = A^2$ and the identity (40) to reach the compact form (42).
5. Ensemble-averaging and expanding in a Pauli basis to obtain (22) with Γ_{ab} given by (45).

14 Google's tech details in RL interface

This section summarizes the RL data interface used in *Reinforcement learning control of quantum error correction* (Google Quantum AI and collaborators, Dec. 8, 2025), aligned with the (a_t, s_t, r_t) notation.

Action a_t

At learning epoch t, the agent samples full control-policy vectors from a Gaussian distribution over all learnable control parameters. A sampled vector is applied to the classical controller and kept fixed while collecting a batch of QEC shots/cycles. Therefore a_t is not a per-gate online action; it is a full parameter setting for one candidate policy.

Observation s_t

The direct environment feedback is detector data from QEC:

- binary detector events (parity flips),
- aggregated into detector-wise detection rates (DR) over finite shots/cycles.

Operationally, for candidate k in epoch t, one can view

$$s_{t,k} = \big(\widehat{\mathrm{DR}}_{t,k,1}, \ldots, \widehat{\mathrm{DR}}_{t,k,m}\big),$$

where m is the number of detector groups (after optional time-translation grouping).

Important detail: in this implementation the policy is mainly parameter-exploring (distribution update over control vectors), not an explicitly state-conditioned neural policy of the form $a_t = \pi_\theta(s_t)$.

Reward r_t from s_t

Google optimizes a surrogate objective based on detector statistics:

$$C = \hat{\mathbb{E}}[D],$$

the average detection-event rate (or equivalently a multi-objective detector-wise form). Lower detector-event rates imply better reward. A consistent scalar choice is

$$r_{t,k} = -C_{t,k},$$

and in multi-objective form one can use $r_{t,k,i} = -\widehat{\mathrm{DR}}_{t,k,i}$ with factor-graph-aware gradient masking.

Decoded logical error rate (LER) is used periodically for evaluation/benchmarking; the main training signal is from detector events and DR.

Are Steane stabilizer histories sufficient to reproduce this flow?

Yes, in principle: full stabilizer measurement histories are sufficient up to deterministic transformations.

Given history data plus schedule metadata, detector-event streams can be constructed by parity/XOR rules (e.g., time-difference events and boundary detectors), then aggregated into DR statistics used as s_t and into reward via $r = -C$.

For practical equivalence to the Google flow, the following pieces are needed:

1. A complete detector definition for the specific circuit (including boundary/time-edge detectors).
2. Transformation from raw stabilizer measurements to detector events.
3. DR estimation over many shots/cycles per candidate policy.
4. Optional grouping of time-translation-equivalent detectors to keep reward dimension $O(d^2)$ for memory experiments.
5. Detector-to-control-parameter mapping (factor-graph sparsity) for efficient gradient estimation.

Hence, raw stabilizer histories are the right primitive data; the missing layer is the detector construction and aggregation pipeline.